

TourSafe

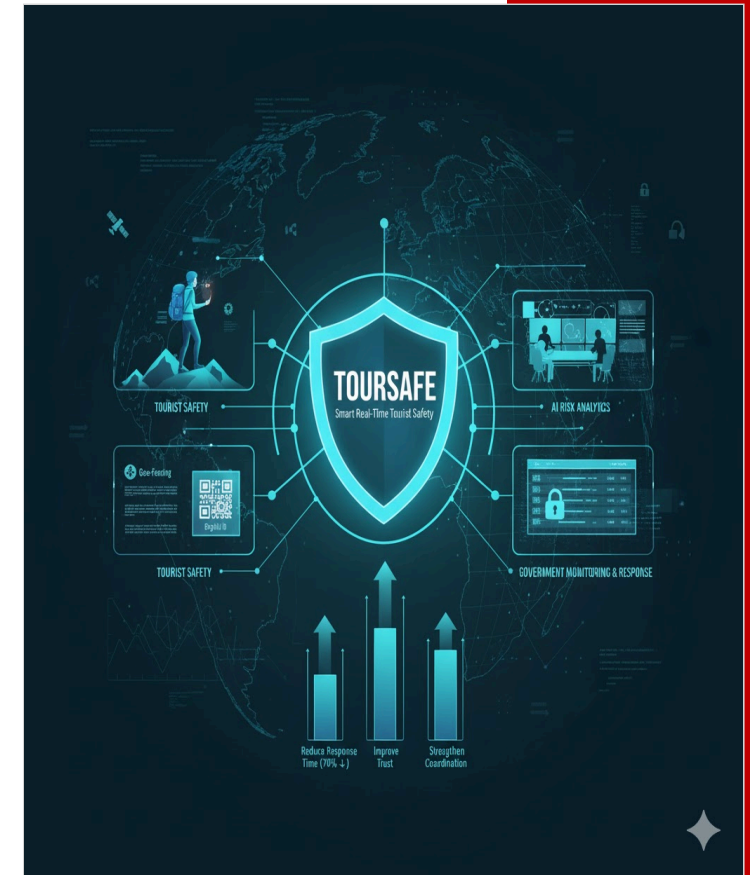
Venture Viability Analysis

SRI SAIRAM ENGINEERING COLLEGE, Chennai

TourSafe

Smart Real Time Tourist Safety Monitoring & Emergency Response System using AI, Geo-Fencing.

SRI SAIRAM ENGINEERING COLLEGE, Chennai



Context

Safety concerns cause drop in tourist arrivals by up to 15%, response time averages 30 - 40 minutes.



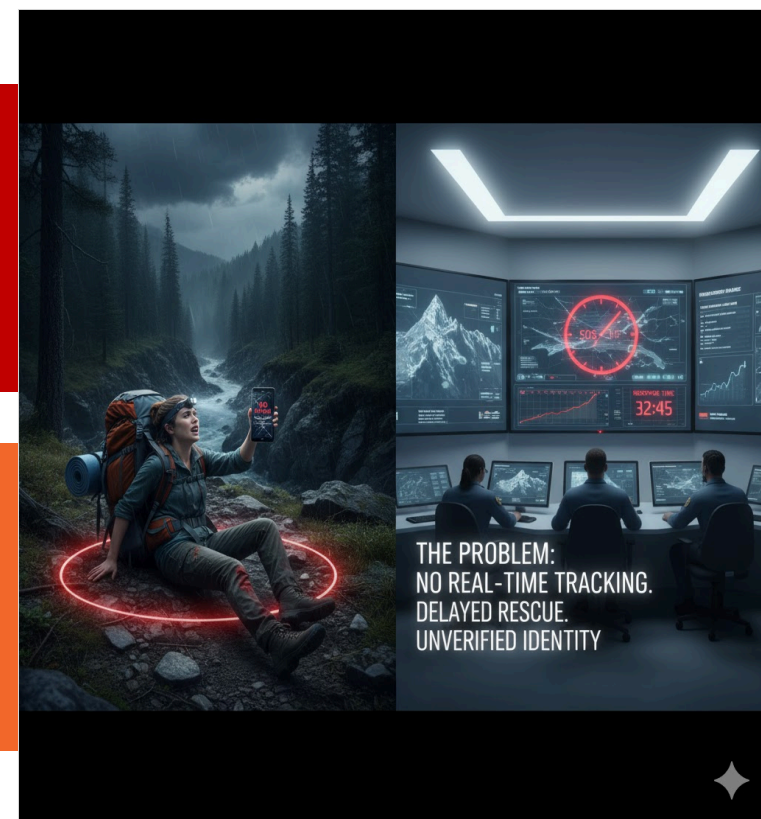
Problem Statement

Problem

Tourist safety is compromised by 30-minute emergency delays in remote areas, zero real-time tracking, and slow, unreliable ID verification.

Impact

Tourists face fatal delays and fear. Authorities battle strained resources and slow ID checks. The industry suffers reputation damage and economic loss



Problem Statement/Industry

Problem Being Solved

Tourists face safety issues in remote or high-risk areas, especially in forests, mountains, or politically sensitive regions where network connectivity and traditional monitoring fail. There is No real-time tourist tracking , Slow emergency response, Difficulty in identity verification, Restricted area violations, Lack of predictive safety alerts

Supporting Data

Safety concerns cause drop in tourist arrivals by up to 15%, response time averages 30 - 40 minutes.

Source: UNWTO Report on Tourist Safety and Travel Trends in India, 2024



Area
Hospitality



Industry
Tourism & -Hospitality



Domain
Travel & Tourism

Problem Analysis



Affected Stakeholders

Tourists visiting remote or high-risk areas face safety risks due to poor monitoring and delayed emergency response. Local authorities, tourism departments, and law enforcement struggle to track, verify, and assist tourists efficiently. This affects tourist trust, regional tourism revenue, and the overall image of travel safety in India.



Impact on Stakeholders

Tourist fatalities stem from 30+ min EMS delays in remote areas and no alerts for restricted zones. This risk fuels tourist fear, degrading the travel experience. Authorities battle resource strain and slower rescue due to lack of real-time tracking/manual reporting. The tourism sector faces reputation damage, lost revenue, and higher insurance premiums.



Root Causes

The problem persists due to fragmented, legacy systems for identification (no unified digital ID/KYC). Poor network coverage in high-risk zones constantly cuts off standard GPS tracking. There is a critical lack of AI-based predictive analytics and proactive geo-fencing capability, meaning emergency response remains severely reactive and too slow.



Personal/Team Connect

We personally felt the fear of isolation during travel. Our passion is deploying AI/TensorFlow skills to move beyond simple tracking and generate instant, life-saving anomaly alerts. We couple this with a Blockchain ID to forge a chain of secure, trusted, real-time help. Our goal: cut emergency response time by 70%, making safety a guaranteed real

Target Customer Segments

Primary

All international and domestic tourists traveling to unfamiliar, high-risk or remote regions.



Secondary

Tourist agencies and government tourism departments.

Customer Segment & Persona

Primary Segment

All international and domestic tourists traveling to unfamiliar, high-risk or remote regions.

Secondary Segment

Tourist agencies and government tourism departments.

Persona



Tourists

Age in years: 28

Location: Suburban

**Organizational Role: {Persona's
primary role}
(if applicable)**

Customer Profile

Education: College pass out

Gender: Female

Occupation: Works for a private company

Interests/Hobbies: Travel

Primary Source of Information: Social Media

Shopping Preference: Hybrid

Comfort with Technology: Medium

Favourite Social Media: Instagram, Twitter, Youtube, Blogs.

Favourite Offline Gathering Spots: Tourist spots

Jobs-to-be-Done

Functional JTBD



The current system limits tourists from key tasks: they can't travel freely without anxiety or ensure rapid rescue, as help often takes 30+ minutes in remote areas. They struggle to validate identity when papers are lost and can't avoid dangerous zones due to lack of proactive, automated geo-fencing alerts and real-time safety intelligence.

Emotional JTBD



Tourists seek to feel secure and trusting in a system powered by Blockchain ID, removing the fear of being ignored or unrecognized. They long to feel relieved and worry-free, without the anxiety that a remote mishap could lead to fatal delay. Above all, they wish to feel empowered, with instant control over their safety, not helpless or uncertain.

Social JTBD



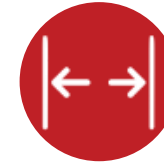
Tourists want to be recognized as responsible travelers, not seen as reckless risk-takers who burden local resources. They seek respect from authorities through verified Blockchain IDs that make their presence trusted. They desire influence by having safety prioritized and being valued as contributors to safe, sustainable tourism, not mere statistics.

Current Alternatives



Current Alternatives

Tourists currently rely on several manual, reactive, and often insufficient solutions: they purchase expensive travel insurance that relies on slow response; they hire local guides whose job is not real-time monitoring; they carry physical documents for slow verification; and they rely on standard GPS and phones, hoping for signal in remote areas.



Gaps in Current Alternatives

Current limitations are severe: costly insurance and guides lack real-time tracking, offering only delayed help. Physical documents are easily lost and fail cross-border verification. Tourists feel unsafe as these systems miss the critical time factor—no instant AI alerts or 10-minute rescue guarantee—leaving them anxious and vulnerable.

Problem Validation (GOOTB)

Partial List of Potential Customers/Users Interviewed

Name: Riya

Occupation: Trekker

Name: john

Occupation: Hiker

Name: Kenzy

Occupation: Cave diver

Problem Validation

Total customers/users interviewed:

- In-person: 5
- Virtually: 5

Total customers/users for whom this problem is important to solve: 6

Total customers/users who are dissatisfied with the current alternatives: 9

Solution Design



Our Solution

TourSafe–Smart Real-Time Tourist Safety Monitoring&Emergency Response System using AI-based Anomaly Detection,Geo-Fencing, and Blockchain-based Digital ID for secure identity and rapid rescue.



Key Features

Our Safety app will have Blockchain-based Digital ID,Geo-fencing alerts for restricted zones, SOS emergency features, AI anomaly dashboard for unusual behavior, Authority dashboard for quick response.



Uniqueness

This solution uniquely combines AI-driven Anomaly Detection with Blockchain-secured Digital ID and real-time Geo-Fencing, delivering predictive safety analytics beyond reactive systems.

Solution Format:

Digital Product - TourSafe : Tourist Safety App

Core Technologies/ Methodologies:

Our Tourist Safety app will be build using Mern Stack, flutter, Figma, React.

Solution Benefits



Functional Benefits

Ensure personal safety through real-time tracking, get instant SOS support, access verified digital ID, receive early risk alerts, and travel confidently with AI-driven protection and secure data privacy



Emotional Benefits

Users gain peace of mind with continuous monitoring and protection, easing anxiety about remote accidents and empowering them with immediate control, turning helplessness into confidence.



Social Benefits

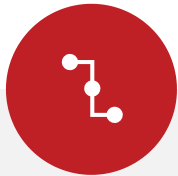
Gain trust and admiration for choosing safe, smart travel, inspire others to explore confidently, earn respect for responsible tourism, and enjoy peace of mind while sharing secure and positive travel



Macro Benefits

TourSafe improves tourist safety through real-time alerts, reducing accidents and saving lives. It strengthens tourism's contribution to the economy, promotes data security and responsible travel.

Competitors



Direct **Competitors**

HollieGuard, TravelSafe



Indirect **Competitors**

Garmin inReach, Spot,
Life360



Direct **Competitors** **Globally**

Everbridge, International
SOS



Indirect **Competitors** **Globally**

Google Maps (offline),
Travel Insurance
providers (e.g., Allianz
Travel), Airbnb

Macro Analysis

Favourable Trends

AREA	DESCRIPTION
Technology	AI and IoT adoption enables real-time tourist tracking and predictive safety alerts.
Economy	Rising travel spending and tourism recovery boost demand for safe, convenient travel apps.

Unfavourable Trends

AREA	DESCRIPTION
Climate/Environment	Dense forests,deep valleys,or extreme weather conditions may interfere with GPS and mobile signals
Legal	Authorities relying solely on the system may overlook the need for manual help.

Data Sources:

Information is based on UNWTO reports, government tourism guidelines, AI & blockchain research, tech docs (Google Maps, Firebase), and news on tourist safety incidents.

Back-of-the-Envelope Financial Projections



Currency: Indian Rupee (INR)

Chosen Business Model: Service Provider

AREA	YEAR 1	YEAR 2	YEAR 3
Revenues	100000000	275000000	630000000
Total Expenses	72700000	104140000	143232000
Profit	27300000	170860000	486768000